

Data Analysis of Road Traffic Accidents to Minimize the rate of Accidents

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Abstract— There has been an influx over the years in people buying cars which has attributed to an increase in the rate of accidents leading to grave losses. This has become a major issue that needs to be dealt with and acknowledged by traffic related law enforcement organizations. The paper deals with an analysis performed which gives greater insights into the causes and factors affecting accidents. Data sets containing more than a million data points each from various regions have been considered. The datasets consist of various features including city names, the type of accident, condition of light, severity, speed zone, consumption of alcohol, time and day of the accident etc. The independent data from these datasets are used to uncover patterns and trends to draw helpful conclusions. These outcomes can be used by traffic-related law enforcement organizations who can take appropriate measures which will result in reducing the risk of loss of lives and occurrence of accidents on roads.

Keywords—Decision trees, python, data analysis, k-means clustering, severity, accidents (key words)

I. INTRODUCTION

Everyday there are large number of road accidents that occur due to various factors that directly or indirectly affect conditions on the road for drivers, passengers and pedestrians alike. These accidents occur due to a myriad of reasons, environmental factors or conditions that have ill effects on road transportation. Several challenges have been faced in this analysis, the primary one being dearth of curated accident datasets. Therefore, researchers specialists utilize regular information mining and prediction models on the accident related data available. Performing trend analysis and recognizing particular patterns might evolve the responsiveness to accidents in metropolitan areas. The dataset consists of attributes such as the city names, the type of accident, the date, time of the accident, condition of light, severity, speed zone, whether it was caused due to the consumption of alcohol, whether it is a hit and run case and so on. In this study, we aim to find common causes of accidents by mining the past accident data. The approach to this study places a focus on both quantitative and qualitative research. Various data mining, visualization techniques have been used

in this study. Statistical analysis using algorithms such as k-means and decision trees to arrive at a substantial value of information. The analysis in this study has been done using Python and R. It is important in a dataset like one involving accidents to be able to pre-process the data. This includes dealing with missing data and noise present in the data. The purpose of this paper is to analyze and contemplate on the solution to accidents and accident response in urban areas.

II. LITERATURE SURVEY

The rapid increment in number of vehicles every year among various countries in the world has additionally enlarged the number of accidents and the rate of the accidents. To analyze the explanation for such an enormous increment in the number of accidents, a few studies have been done to determine the reasons behind them, thereby attempting to recognize approaches to forestall and deter the rate of accidents. A portion of those studies utilizing distinctive strategies for machine learning, data mining have been examined to help in its understanding.

S Erdogan et al. [1] used Geographical Information System (GIS) technology for visualizing accident data and analysis in highways for hot spots had been carried out. These hot spots were investigated and decided using two different methods of repeatability analysis and Kernel Density analysis, which resulted in finding out problematic location such as junctions, cross roads etc. Mohamed A Abdel-Aty et al. [2] developed a model using Negative Binomial modelling technique to estimate the accident frequency occurrence and involvement. The model showed the importance of the Annual Average Daily Traffic, lane, shoulder, and degree of the horizontal curvature on the frequency of accident. Another study showed the prospect of using Bayesian Networks to classify road traffic accidents according to the severity of injury. Juan Mujalli et al.[3] had a model based on BNs are able to make predictions not having the necessity of pre-assumptions. BNs are used to classify these accidents according to the severity of injury and also is involved in identifying variables correlated with severely injured or killed based on type of accident, age,

lightening and number of injuries. An analysis on the impact of collision by using three-dimensional technologies was done by Ursula Buck & Silvio Naether [4]. They use documentation of the external findings of a body and to inflict injuries. This also involves 3D surface scanning. The study also included 3D models processing and resolution of the driving direction of the vehicle, geometric determination of the impact situation.

Ju-Yeon Lee et al. [5] expressed accident size as the number of vehicles involved, the number of vehicles damaged, and the number of injured or the number of deaths. A structural equation model was used to understand complex relationships among variables. Miao M. Chong et al. [7] used decision trees and ANNs (Artificial Neural Networks) to model the severity of injuries resulting from traffic accidents and to gain information about the most important factors in fatal injuries. The study displays that there are three essential attributes in fatal injuries. They are light condition of the roadway, driver's seat belt usage and driver's alcohol usage. Miaou SP [8] in 1994 dealt with the relationship between the geometric design of certain sections of roads and accidents of trucks. There were three different models used: Poisson regression, NB regression and zero-inflated Poisson (ZIP) regression. Maher MJ et al. [10] worked on an extensive methodology for the development of predictive accident models. Another idea that was worked on upon was identifying the key risk factors of injury severity. Rovsek V et al. [11] used a non-parametric classification tree. It is key to establish the most essential patterns and attributes that significantly impact accident severity and injuries. There have also been multiple regional research that's been done. That is, macro analysis of an area. Annisa Jusuf et al. [12] worked on traffic accidents in Indonesia and derived insights pertinent to regions. Another similar one is by Pardillo-Mayora JM et al. [16] where roadside hazardousness index was a metric that was used and Spanish rural lane roads were analysed. There are also other psychological aspects that have been researched. Dianne Parker et al. [6] have studied behavioural characteristics and involvement of different types of accidents.

Most of the studies that were concluded on road traffic accidents using different models and ways of machine learning and analysis found out that root causes of the accidents are due to driver's behavior, light conditions and alcohol consumption by drivers and road conditions. These are the major attributes that contributes most to the increasing rate of accidents and identified several hot-spot zones where density of occurrence of accidents and severely high and results in high number of fatalities.

III. DESIGN

The pre-processing of data involves analyzing types of attributes and how important each feature is to our analysis. The system is designed as any basic machine learning model which extracts contributing features, scales the features in the

pre-processing stage and then subsequently trains and tests the model to obtain appropriate analytical results.

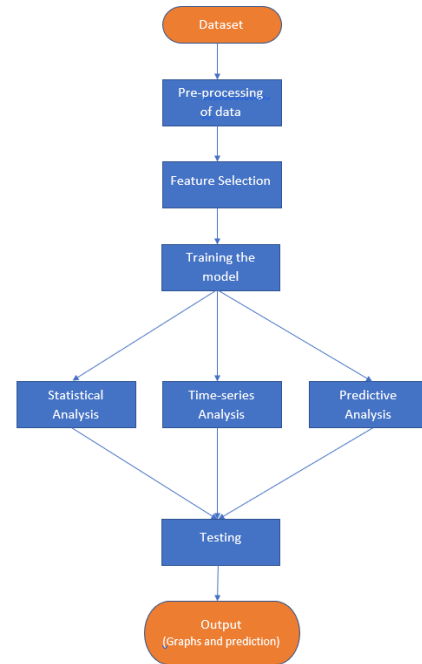


Fig 1. Architecture and Design of the system

IV. IMPLEMENTATION

The statistical analysis and the data visualization were done using Python primarily and also R. Some of the open source libraries that were used are listed below:

- A. **scikit-learn:** Scikit-learn is a free software machine learning library for the Python programming language. It features simple and efficient tools for data analysis and data mining.
- B. **Matplotlib:** Matplotlib is a 2D plotting library for the Python programming language and its numerical mathematics extension NumPy.
- C. **Seaborn:** Seaborn [18] is a Python visualization library based on matplotlib. It provides a high-level interface for drawing attractive statistical graphics.
- D. **Party:** Used for recursive partitioning. The crux of the package is ctree, which is an implementation of conditional inference trees [19] which embed tree-structured regression models.

Decision tree develop regression or classification models in the form of a structure of a tree. It breaks down a dataset into smaller subsets. At the same time an associated decision tree is incrementally developed. The final result is a tree

with decision nodes. We need to calculate homogeneity of samples and the algorithm uses entropy to do the same. The entropy is zero, then the sample would be completely homogenous. If the sample is an equally divided it has entropy of one. The entropy [13] of the sample is acquired using the formula (1), where $H(S)$ denotes entropy, $p(x)$ is the number of successful cases.

$$H(S) = - \sum_{x \in X} p(x) \log_2 p(x) \quad (1)$$

Decision tree was used to classify the severity of accidents. Various factors like speed zones, light conditions, etc. were considered and fed into the decision tree classifier.

The dataset is then split on the different attributes. The entropy for each branch is calculated. At that point it is included relatively, to get the total entropy for the split. The subsequent entropy is subtracted from the entropy before the split. The outcome is the Information Gain or reduction in entropy. This is achieved using the formula (2), where IG denotes Information Gain, H gives the entropy and p is the number of successful cases present for each case.

$$IG(A, S) = H(S) - \sum_{t \in T} p(t) H(t) \quad (2)$$

Algorithm:

- Read the dataset
- Split the training set into a subset of different attributes and pick the attribute at the root of the tree.
- Calculate entropy and information gain values and pick the highest gain attribute.
- Repeat the same thing in step 3 for sub-trees till the final tree is obtained.

K-means clustering is a method of vector quantization, initially from signal processing, that is popular for cluster analysis in data mining. *K-means* clustering [22] looks to to partition n observations into k clusters. Each observation belongs to the cluster with the nearest mean, serving as a prototype of the cluster. If we are given a set of observations $(x_1, x_2, \dots, x_n)$, where each observation is a d -dimensional real vector, then *k-means* clustering aims to split the n observations into k such that k is less than or equal to n ($k \leq n$) and here, μ_i is the mean of points in S_i .

$$\arg \min \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2 \quad (3)$$

The determination of the number of clusters [19] in *k-means* is difficult. So, we used the Rule of thumb which sets the number of clusters as below

$$k \approx \sqrt{\frac{n}{2}} \quad (4)$$

Here n is the number of the objects or the data points.

Algorithm:

- Read the dataset and have a set V which is a set of 'k' centers.
- Select random 'k' clusters
- The distance between cluster centers and each data point is to be calculated.
- The data point to the cluster center is to be assigned. Its distance from the cluster center is minimum of the centers.
- Recalculate the new cluster using the formula:

$$\mu_i = \left(\frac{1}{c_i} \right) \sum_{j=1}^{c_i} x_j$$

where, ' c_i ' denotes the number of data points in i^{th} cluster.

- The distance between new obtained clusters and each data point is to be recalculated.
- If no data point was reassigned then stop, else repeat instructions from step (iv).

V. RESULTS

A. Peak time of the accidents

Using the time of the day attribute, time series analysis was performed to obtain the peak time accidents during the day.

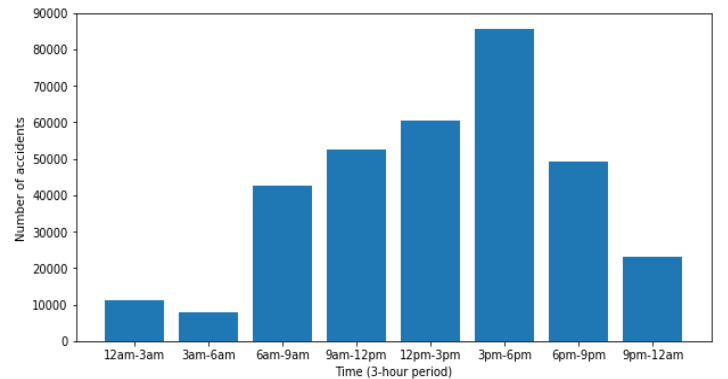


Fig. 2. Accidents during the day with 3-hour time period

The plot with accidents in 3-hour periods, (Fig. 2) of the dataset gives us a peak in 3pm-6pm. This peak could be attributed to the fact that most vehicle traffic is post working hours. Many of the accidents are around a working area and thus, many accidents happen when the working class is done for the day and is heading home.

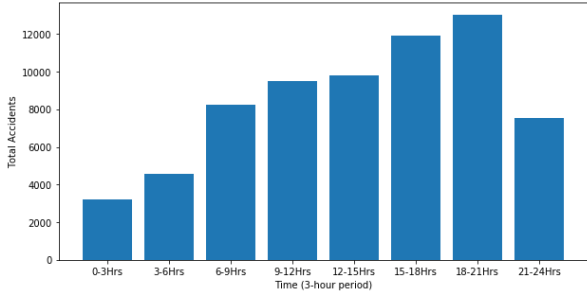


Fig. 3. Comparison of total accidents of Karnataka, Andhra Pradesh and Tamil Nadu with 3-hour time periods

The bar graph in Fig 3. helps us in understanding of trends of accidents in various states in India. Karnataka, Andhra Pradesh and Tamil Nadu are 3 states that have been considered and it can be seen that the trends are between 3pm-9pm, but the highest density is in the 6pm-9pm period. The peak hours in the second plot are different to that in the first.

B. Vehicle Type and Faults

A dataset was considered to analyze the types of vehicles involved in accidents. As shown in Fig. 4, the roles different vehicles play in accidents can be observed. The highest percentage, which is around 35% is to 2-wheelers. The percentage of heavy vehicles which include trucks, heavy duty lorries, tractors, etc. is also reasonably high.

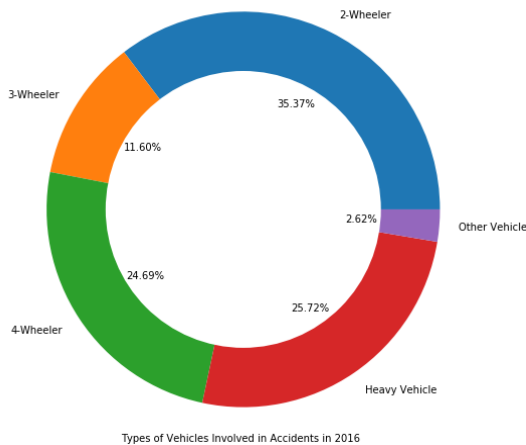


Fig. 4. Types of vehicles involved in 2016

Another interesting factor is to consider whose fault it is for these accidents to have taken place. The chart below in Fig. 5, tells us that it is the driver's fault over 80% of the times. Contrary to popular belief, the fault does fall on the driver in most cases. It is also important to note other factors like stray

animals, light, road and weather conditions, etc. also play a part in these accidents taking place.

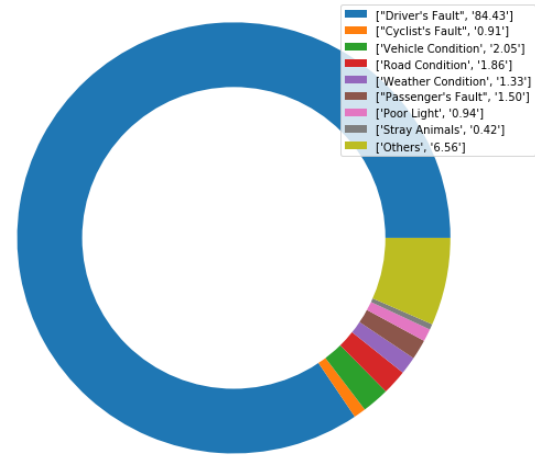


Fig. 5. Percentage of accident due to fault of several road users

C. Accident Severity Index

This is a metric which is defined as total deaths per 100 accidents. This factor gives us information about accidents involving fatalities. There is an interesting trend that has been observed. The number of vehicles have always been increasing every year, and this could also be a cause in the trend observed in Fig. 6. A small rise can be seen in the years 2012 and 2014. But, a great rise is seen from 2014 to 2016. This implies, an increase in the fatalities per 100 accidents in the 2 years. With vehicles increasing year by year, this inference is something to be looked into. Solutions to this could be improving ambulance facilities and emergency services so as to reach the ones severely injured in the accidents better.

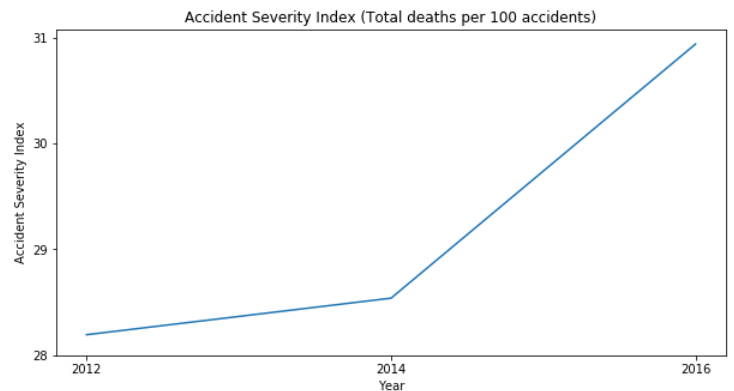


Fig. 6. Accident Severity Index

D. Day wise alcohol consumption

Through various tests and plots, we find that the consumption of alcohol plays a major role in accidents. The pie chart shown below in Fig. 7, shows us that over 61% of the accidents happen over the weekend (Friday-Sunday). This is a major concern and stricter measures of security, increase in police personnel could be two solutions that enforcement agencies can look into in areas of pubs, bars and restaurants over the

weekend, so as to attempt in minimizing the rate of accidents. These measures could play a huge role in dealing with this type of accidents.

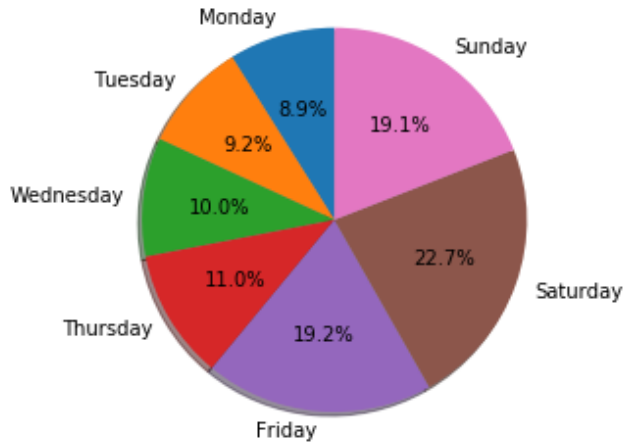


Fig. 7. Percentage of accident due to alcohol consumption on different days of week

E. Decision Tree

A decision tree was used to classify accidents as more severe or less severe. The tree below (Fig. 8) takes into account features like speed zone, alcohol consumption and light conditions. The tree's root node is speed zone. From the decision tree, we infer accidents which happen in speed zones > 80 kmph, are prone to be more severe. The other factors which play a major role in accidents where speed zone is lesser than 80 kmph, are consumption of alcohol and light conditions.

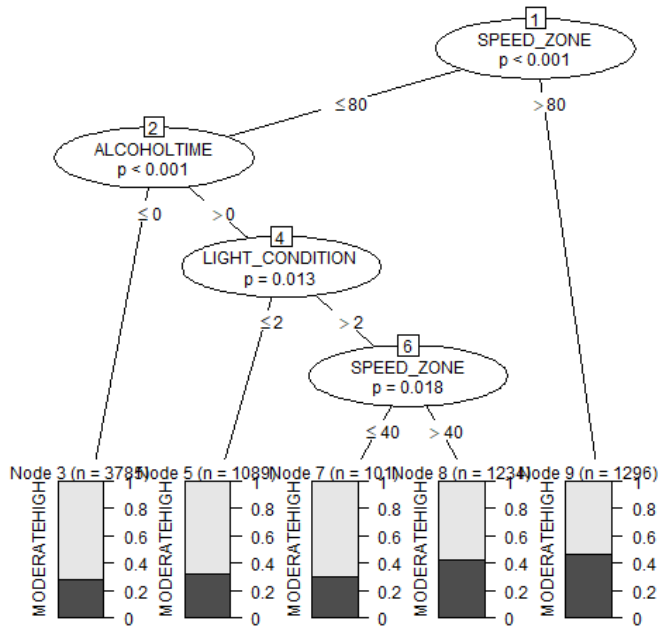


Fig. 8. Decision tree for severity of accidents

F. Clustering of regions

The K-means clustering algorithm is implemented to cluster regions in which accidents have happened to find 'hotspot' zones of accidents. As shown in Fig. 9, each of the cities in the country is associated with an integer value in the range of 1 to 25. It was concluded that the region of cities belonging in the range of 20-25 experienced a higher density of accidents as the clusters are more here and the cities numbered as 4,9,13 and 16 experienced comparatively lesser number of accidents leading to death of people.

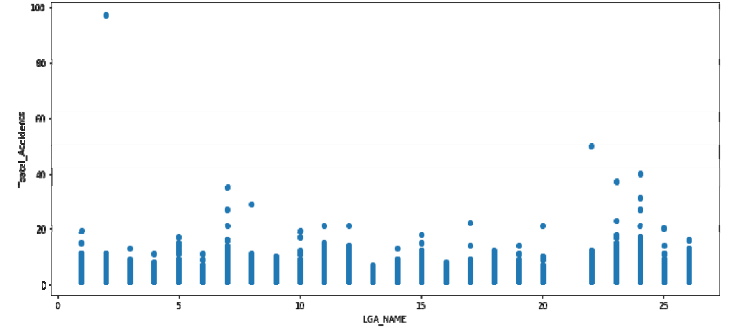


Fig. 9. Total accidents clustered into different regions

G. Severity Analysis based on Light Conditions, Speed Zone and Alcohol consumption

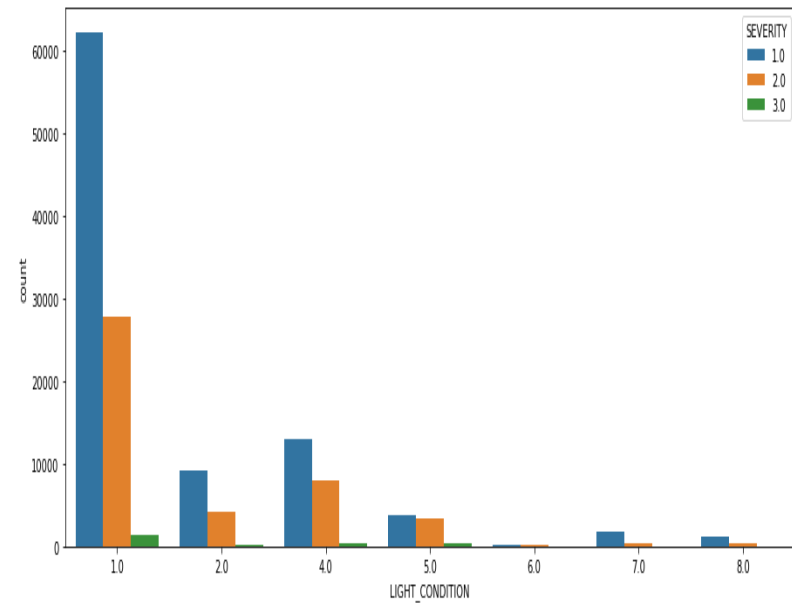


Fig. 10. Count plot of light condition, severity with total accident

From the count plot shown in Fig. 10, we can see how the light conditions, 1 being least and 8 being brightest light conditions, we compare how severity of accidents are affected by the light conditions. It is seen that in places with average or mediocre light, which is in the range of 2-5, accidents are higher with the consumption of alcohol. This could be due to the fact that in lack of proper light, the decision of a driver with the consumption of alcohol gets even more clouded.

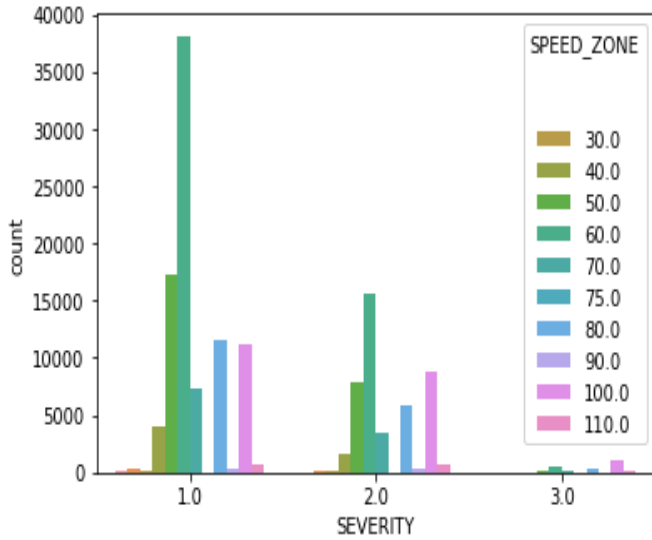


Fig. 11. Count plot of speed zone and severity

The count plot in Fig. 11, describes how speed zone of the area in which the accident took place affects the severity of the accident. The most severe accidents (with severity=1), show spikes primarily in the 60kmph region. This shows that areas with those speed zones need to be dealt with more caution. Increase in safety measures in such regions could be a probable measure to deal with the situation.

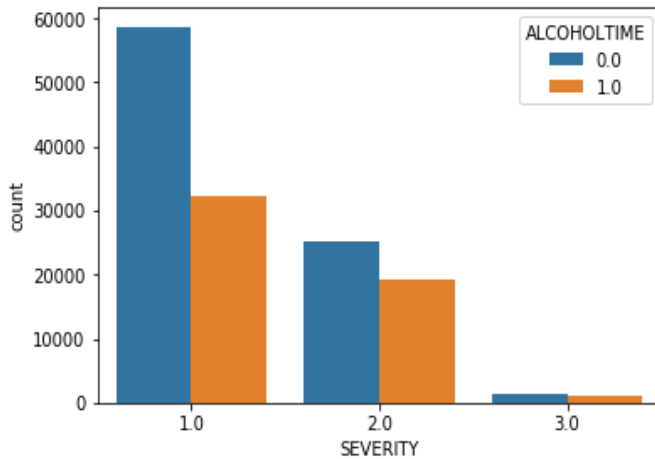


Fig. 12. Count plot of severity and alcohol consumption

The severity of alcohol related accidents averaged around 40,000 and it drastically reduced when there was no alcohol consumed by the injured party. This leads to the conclusion that the majority of the situations when the driver of the vehicle has consumed alcohol, the severity was more, which

would have caused an accident. The bar graph for this analysis is depicted in Fig. 12.

VI. CONCLUSION

Through the analysis, various inferences and solutions to the same were achieved. Around 61% of accidents involving alcohol happen on the weekend. This is a major concern and stricter measures of security, increase in police personnel could be two solutions that enforcement agencies can look into in areas of pubs, bars and restaurants over the weekend, so as to attempt in minimizing the rate of accidents. With the peak time of accidents obtained, there should be proactive measures taken in these specific regions so as to deal with these issues more effectively. For the dataset taken, it was observed that regions 20-25 experienced higher density of accidents. Extra caution can be taken in these 'hotspots'. Contrary to popular belief, it is observed that it is the driver's fault in over 80% of the accidents. This could be due to the carelessness and recklessness of the drivers. The govt. could always invest on having awareness campaigns of road safety throughout the year to educate the people. The accident severity index (deaths per 100 accidents) has shown an increasing trend from 2012-2016. This implies that fatalities on the road are increasing year by year. Solutions to this could be improving ambulance facilities and emergency services so as to reach the ones severely injured in the accidents better. Various types of vehicles pose different problems on the road. The percentage of accidents involving heavy duty vehicles is a substantial number compared to the number of those vehicle on the road. This could mean multiple lanes for different types of vehicles to deal with accidents better.

REFERENCES

- [1] S Erdogan, I Yilmaz, T Baybura, M Gullu - Geographical information systems aided traffic accident analysis system case study, 2008
- [2] Mohamed A Abdel-Aty, A Essam Radwan - Modeling traffic accident occurrence and involvement, 2000
- [3] de Ona, Juan Mujalli, Randa Oqab Calvo, Francisco J - Analysis of traffic accident injury severity on Spanish rural highways using Bayesian networks, 2011
- [4] Ursula Buck & Silvio Naether - 3D Accident Reconstruction: New Measuring Techniques for 3D Documentation of Crashed Cars, 2007
- [5] Ju-YeonLee, Jin-HyukChung, BongsooSon - Analysis of traffic accident size for Korean highway using structural equation models, 2008
- [6] DianneParker, RobertWest, SteveStradling, Antony S.R.Manstead - Behavioural characteristics and involvement in different types of traffic accident, 1995
- [7] Miao M. Chong, Ajith Abraham & Marcin Paprzycki - Accident Analysis Using Decision Trees and Neural Networks, 2004
- [8] Miaou SP. The relationship between truck accidents and geometric design of road sections—poisson versus negative binomial regressions, 1994
- [9] Modeling vehicle accidents and highway geometric design relationships, accident analysis and prevention - Miaou SP, Lum H. 1993.
- [10] Maher MJ, Summersgill I. A comprehensive methodology for the fitting of predictive accident models. Accid Anal Prev Elsevier. 1996;

- [11] Identifying the key risk factors of traffic accident injury severity on Slovenian roads using a non-parametric classification tree, transport. UK: Taylor and Francis - Rovsek V, Batista M, Bogunovic B. 2014.
- [12] Annisa Jusuf, Ignatius Pulung Nuraprasetyo and Arya Prihutama : Macro Data Analysis of Traffic Accidents in Indonesia, 2017
- [13] The Base Strategy for ID3 Algorithm of Data Mining Using Havrda and Charvat Entropy Based on Decision Tree - Nishant Mathur, Sumit Kumar, Santosh Kumar, and Rajni Jindal.
- [14] Chen Chen : Analysis and Forecast of Traffic Accident Big Data, 2017.
- [15] Jones B, Janssen L, Mannering F. Analysis of the frequency and duration of freeway accidents in Seattle, accident analysis and prevention. Elsevier; 1991
- [16] Pardillo-Mayora JM, Domínguez-Lira CA, Jurado-Pina R. Empirical calibration of a roadside hazardousness index for Spanish two-lane rural roads. *Accid Anal Prev*. 2010.
- [17] Kashani T, Mohaymany AS, Rajbari A. A data mining approach to identify key factors of traffic injury severity, *promet-traffic & transportation*, vol. 23; 2011.
- [18] Seaborn visualisation : <https://seaborn.pydata.org/>
- [19] Ctree package:
<https://www.rdocumentation.org/packages/partykit/versions/1.2-1/topics/ctree>
- [20] glaros.dtc.umn.edu/gkhome/fetch/papers/vaclusterDMKD05.pdf
- [21] Fraley C, Raftery AE. How many clusters? Which clustering method? Answers via model-based cluster analysis. *Comput J*. 1998
- [22] K-modes clustering. *J Classif*, Chaturvedi A, Green P, Carroll J. 2001
- [23] Exploratory latent structure analysis using both identifiable and unidentifiable models by Goodman LA. 1974
- [24] Australian AID: Indonesia Infrastructure Initiative, Modernizing the National Road Network: A Planning Framework to Improve Connectivity and Development, Consultant Report, 2012.
- [25] DianneParker, RobertWest, SteveStradling, Antony S.R.Manstead - Behavioural characteristics and involvement in different types of traffic accident, 1995